



Alberta Public Health Disease Management Guidelines

Gonorrhea



This publication is issued under the Open Government Licence – Alberta (<http://open.alberta.ca/licence>). Please note that the terms of this licence do not apply to any third-party materials included in this publication.

This publication is available online at <https://open.alberta.ca/publications/gonorrhea>

For further information on the use of this guideline contact:

Health.CD@gov.ab.ca

Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

Gonorrhea | Alberta Health, Government of Alberta

© 2021 Government of Alberta | October 2021

Revisions

Revision Date	Document Section	Description of Revision
October 2021	General	• Updated Template • Diagnosis and Treatment section moved to Epidemiology • Updated web links and added where applicable
	Incidence	• Removed graphs as outdated

Table of Contents

Revisions	3
Case Definition	5
Confirmed Case	5
Reporting Requirements	6
Physicians, Health Practitioners and Others	6
Laboratories.....	6
Alberta Health Services - STICS.....	6
Additional Reporting Requirements for Physicians, Health Practitioners and Others	7
Clinical Presentation and Epidemiology	8
Etiology	8
Clinical Presentation.....	8
Diagnosis.....	8
Treatment.....	9
Reservoir.....	9
Transmission.....	9
Incubation Period	10
Period of Communicability	10
Host Susceptibility	10
Incidence	10
Public Health Management	11
Key Investigation	11
Management of a Case.....	11
Management of Contacts.....	12
Preventive Measures.....	12
Screening	13
References.....	14

Case Definition

Confirmed Case

Footnote ^(A) applies to this section.

Genital and Extra-Genital Infections

Laboratory evidence of infection in genitourinary specimens^(B) or from pharynx, rectum, joint, conjunctiva, blood or other extra-genital sites:

- Isolation of *Neisseria gonorrhoeae* by culture

OR

- Molecular detection of *N. gonorrhoeae* (e.g., Nucleic Acid Amplification Testing [NAAT])

Perinatally Acquired Infections

Laboratory evidence of infection leading to the diagnosis of gonococcal conjunctivitis, scalp abscesses, vaginitis, bacteremia, arthritis, meningitis or endocarditis in an infant in the first four weeks of life:

- Isolation of *N. gonorrhoeae* by culture

OR

- Molecular detection of *N. gonorrhoeae* (e.g., NAAT)

^(A) Each case classification is mutually exclusive. Individuals with more than one site of infection concurrently may fall under more than one case classification but will be counted as one case with multiple sites of infection identified to avoid duplicate counting of cases.

^(B) Refer to the [Public Health Laboratory \(ProvLab\) Guide to Services](#) for current specimen collection and submission information.

Reporting Requirements

Physicians, Health Practitioners and Others

Note: This section includes First Nations and Inuit Health Branch

Physicians, health practitioners and others shall notify the Sexually Transmitted Infection (STI) Medical Director^(C) via Sexually Transmitted Infection Centralized Services (STICS), of all confirmed cases within 48 hours (two business days) by forwarding a completed [Notification of STI](#) form.

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- STI Medical Director via STICS, and
- Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services - STICS

Contact Information: Toll free: 1-855-945-6700 option 4
Fax: 780-670-3624

- The STI Medical Director/STICS are responsible for ensuring investigation and follow-up of all reported confirmed cases.
- The STI Medical Director/STICS shall forward the initial Notification of STI form of all confirmed cases to the CMOH (or designate) within two weeks of notification and the final Notification of STI form within four weeks.
- The STI Medical Director/STICS shall forward the [Perinately Acquired Notifiable Disease Enhanced Report](#) form for all confirmed and probable cases of perinately acquired disease to the CMOH (or designate) within four weeks of notification.
- For out-of-province and out-of-country reports, the following information (when available) should be forwarded to the CMOH (or designate) by phone, fax or secure/encrypted electronic transfer as soon as possible:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

^(C) The STI Medical Director is the Provincial Medical Director of Alberta Health Services' Sexually Transmitted Infection Centralized Services (STICS) and is also a Medical Officer of Health.

- For out-of-province and out-of-country contacts the following information (when available) should be forwarded to the CMOH (or designate) as soon as possible:
 - name,
 - date of birth,
 - date of exposure, and
 - out-of-province/country contact information.

Additional Reporting Requirements for Physicians, Health Practitioners and Others

In all cases, where a person under 18 is suspected or confirmed to have an STI, an assessment should be carried out by the clinician to determine if additional reporting is required.

To Alberta Child and Family Services

- The clinician should determine whether there are reasonable and probable grounds to believe that they are in contact with “a child in need of intervention” (as per Section 1(2) of the [Child, Youth and Family Enhancement \(CYFE\) Act](#)) and shall report to a director pursuant to Section 4 of the *CYFE Act*.⁽¹⁾
- Reporting is done by contacting the local Child and Family Services office or calling the **CHILD ABUSE HOTLINE: 1-800-387-5437 (KIDS)**. For local office contact information see [the ministry’s website](#).

To Law Enforcement Agency

- Consent is a key factor in determining whether any form of sexual activity is a criminal offence. Children under 12 do not have the legal capacity to consent to any form of sexual activity. The law recognizes that the age of consent for sexual activity is 16. However, the law identifies the exception for minors between 12 and 16 years as having the ability to consent, in “close in age” or “peer group” situations.⁽²⁾
- Reporting is done by contacting your local City Police Detachment or [RCMP Detachment](#).
- For additional information see:
 - Alberta [Child, Youth and Family Enhancement Act](#)
 - Age of Consent to Sexual Activity at www.justice.gc.ca/eng/rp-pr/other-autre/clp/faq.html
 - Criminal Code of Canada at www.laws-lois.justice.gc.ca/eng/acts/C-46/⁽³⁾

Clinical Presentation and Epidemiology

Etiology

Neisseria gonorrhoea is an aerobic gram negative bacteria with 14 subtypes.⁽⁴⁾

Clinical Presentation

Genital Infections

In men, urethral infection is commonly symptomatic for urethral discharge and/or dysuria. The discharge is often mucopurulent or purulent. Rarely, epididymal tenderness/swelling or balanitis may be present.^(5,6)

Women are often asymptomatic (up to 50%).^(4,7) Commonly, no abnormal findings are present on examination; however, if symptoms are present they may include mucopurulent endocervical discharge and cervical friability.^(5,6)

Extra-Genital Infections

Extra-genital Infections include infection in the pharynx, rectum, joints, conjunctiva, blood, and other sites. Pharyngeal and anorectal infections are common and are often asymptomatic. Anorectal infections may cause pruritus, tenesmus, and discharge. Conjunctivitis may occur, and, if not treated adequately, could cause blindness. Bacteremia is rare, occurring in 0.5–1% of gonorrheal infections. Meningitis, arthritis, skin lesions, and endocarditis also occur infrequently. Death is uncommon.^(5–7)

Perinatally Acquired Infections

Infections occur in newborns as a result of passage through an infected birth canal. The most common presentation of infection is ophthalmia neonatorum. Other presentations such as vaginitis, rhinitis, anorectal infection, funisitis, urethritis, scalp abscesses or other disseminated diseases (bacteremia, arthritis, meningitis or endocarditis) may also occur.^(5,6,8)

Diagnosis

The diagnosis is established by the identification of *N. gonorrhoea* at an infected site. Nucleic acid tests were first introduced in Alberta in 1997 and are preferred in situations where urine testing is more feasible or delays in transport of specimens may occur. As part of Alberta's plan to maintain surveillance for antimicrobial resistant gonorrhea, the following sites or indications are situations where gonorrhea culture, in addition to molecular (NAAT) testing, is routinely performed or recommended:

- infection acquired outside of Alberta,
- treatment failure,
- sexual contact with individual from a country with high prevalence of gonorrhea drug resistance,
- sexual abuse of children,
- sexual assault, and

- symptomatic men who have sex with men (MSM).

Treatment

Indications for treatment are:

- positive diagnostic test result,
- diagnosis of a syndrome compatible with a gonorrheal infection, without waiting for test result, and/or
- partner with a positive gonorrhea test result or diagnosis of a syndrome compatible with a gonorrhea infection in a partner without waiting for test results.

Refer to the current [Alberta Treatment Guidelines for STI in Adolescents and Adults](#).

Considerations

- Disseminated infections and those involving the eye require expert consultation with STICS.
- Available data suggest azithromycin is safe and effective in pregnant women.

Pediatric Cases

- When a case is diagnosed in an infant, the mother and her sexual partner(s) should be examined and tested.
- It is recommended that all children < 14 years of age be referred to a pediatrician and, because of the high risk of sexual abuse (excepting those < one month of age with a conjunctivitis or < six months of age with pneumonia), be managed in consultation with one of the following referral centres.

Edmonton

Child and Adolescent Protection Centre
Stollery Children's Hospital
1C4.24 Mackenzie Health Sciences Centre
8440-112 Street
Edmonton, Alberta T6G 2B7
Tel: 780-407-1240

Calgary

Child Abuse Service
Child Development Centre
Suite 200, 3820-24 Ave NW
Calgary, Alberta. T2N 1N4
Tel: 403-955-5959

Reservoir

The only known reservoir is humans.^(4,6)

Transmission

Neisseria gonorrhoeae is transmitted by direct inoculation of infected secretions from one mucous membrane to another, usually through sexual activity or through the birth process (vertical transmission).^(6,8)

Incubation Period

The incubation period is typically two to seven days, with a range of 1–14 days.⁽⁶⁾

Period of Communicability

Neisseria gonorrhoeae is communicable for as long as the person harbours the organism.

Host Susceptibility

All persons are susceptible to this disease if exposed and re-infection is common. Co-infection with *Chlamydia trachomatis* is common. Epidemiologic studies provide strong evidence that gonorrheal infections facilitate HIV transmission.^(6–8)

Incidence

Genital gonorrhea became reportable in Canada in 1924. Cases and rates in Alberta experienced a steady rise since 1998 with a short period of annual variability during the years 2006–2015. Since 2015, the rate has risen dramatically.

See the [Interactive Health Data Application](#) for more information.

Public Health Management

Key Investigation

The diagnosis and treatment is performed by community health care providers.

- Determine the presence or absence of symptoms.
- Determine if behaviors that increase risk for gonorrhea are present:
 - sexual contact with gonorrhea infected person(s),
 - new sexual partner or more than two sexual partners in preceding year,
 - previous STI, and/or
 - vulnerable populations (e.g., injection drug use (IDU), incarcerated individuals, people involved in exchanging goods for sex, street involved youth, etc.).
- Offer testing for HIV and other STI.
- Counsel and identify partners, including locating information.

Management of a Case

- Test of cure by molecular detection (NAAT) is indicated for all cases (three to four weeks after completion of treatment).
- All patients treated for gonorrhea should also be treated for chlamydia infection regardless of chlamydia test result.
- Immediate treatment is recommended in men and women with suspected urethritis, cervicitis or proctitis.
- Repeat testing for all individuals with gonorrhea infections is recommended six months post- treatment.
- All cases should be instructed about infection transmission. Patients should be counseled about the importance of abstaining from unprotected intercourse until seven days after completion of treatment by both case and partner(s).
- All cases should be provided with individualized STI prevention education, targeted at developing knowledge, skills, attitudes and behaviors to reduce the risk and prevent recurrences of STI.
- Immunization against hepatitis A, B and HPV may be recommended. Refer to the [Alberta Immunization Policy](#) for immunization eligibility.
- All patients with a notifiable STI qualify for provincially funded medications.
 - STICS will send replacement medication upon receipt of a Notification of STI Form when the Health Care Provider mailing address is indicated on the form.
 - Health care providers may order additional quantities of most medications by contacting STICS.

To obtain the phone number for your designated **Partner Notification Nurse**, or for advice on management of your case, call **STICS Toll free: 1-855-945-6700 option 4**

- Sexual assault in adults should be managed in conjunction with local Sexual Assault services and other appropriate community support services.

Recalcitrant Patients

The [Public Health Act](#) (sections 39 through 52) authorizes detention of recalcitrant patients for medical examination, treatment and/or counselling.

- The CMOH (or designate) or MOH (or designate) may issue a certificate to detain an individual who is believed to be infected and refuses or neglects to comply with treatment.
- In order to enact this provision, there must be proof of infection or contact with an infected person and:
 - documentation of failure to comply with prescribed treatment and medical examination, or
 - non-compliance for testing and/or treatment.

Management of Contacts

Partner Notification

- Partner notification will identify those at risk, reduce disease transmission/re-infection and ultimately prevent disease sequelae.
- **It is mandated under the *Communicable Disease Regulation* that every attempt is made to identify, locate, examine and treat partners/contacts of all cases.**
- All contacts should be:
 - screened for HIV and other STI,
 - instructed about infection transmission, and
 - provided with individualized STI prevention education, targeted at developing knowledge, skills, attitudes and behaviors to reduce the risk and prevent recurrences of STI.
- Health care providers are required to provide partner names and locating information on the Notification of STI form and forward to STICS.
- If testing and/or treatment of partner(s) are not confirmed on the Notification of STI form, STICS will initiate follow up by a Partner Notification Nurse (PNN).
 - PNNs are specially trained to conduct notification of partners and contacts in a confidential manner that protects the identity of the index case.

The phone number for your designated PNN is available by calling **STICS Toll free: 1-855-945-6700 option 4**

- STICS will follow-up on any incoming referrals of cases and partner(s) from all out of province/country referrals.

Preventive Measures

- Ensure appropriate treatment of *N. gonorrhoeae* for cases.
- Interview case, identify and ensure appropriate treatment and follow-up of *N. gonorrhoeae* for sexual partner(s).
- Ensure STI care is culturally appropriate, inclusive, readily accessible, and acceptable.
- Include information about risk for STI during pre-travel health counseling.

- Educate the case, sexual partners, and the public on methods of personal protective measures, in particular the correct and consistent use of condoms and discuss safer sex options including:
 - delaying onset of sexual activity,
 - developing mutually monogamous relationships,
 - reducing the numbers of sexual partners, and
 - discouraging behaviors associated with the acquisition and transmission of STI.

Screening

- Individuals with risk factors for gonococcal infections:
 - sexual contact with gonococcal infected person(s),
 - new sexual partner or more than two sexual partners in preceding year,
 - previous STI, and
 - vulnerable populations (e.g., IDU, incarcerated individuals, those who exchange goods/money for sex, and street involved youth).
- All sexually active persons under 25 years of age, at least annually.
- All pregnant women. See [Alberta Prenatal Screening Guidelines for Select Communicable Diseases](#).
- Women prior to insertion of an IUD, a therapeutic abortion, or a dilation and curettage.
- Victims of sexual assault.

References

1. Department of J. Child Youth and Family Enhancement Act Policy Manual. Government of Alberta. Government of Alberta; 2010.
2. Government of Canada. Age of Consent to Sexual Activity [Internet]. Ottawa: Government of Canada; 2011. Available from: <http://www.justice.gc.ca/eng/dept-min/clp/faq.html>
3. Department of Justice. Criminal Code of Canada. Government of Canada. Government of Canada; 2011.
4. Unemo M, Golparian D, Sánchez-Busó L, Grad Y, Jacobsson S, Ohnishi M, et al. The novel 2016 WHO *Neisseria gonorrhoeae* reference strains for global quality assurance of laboratory investigations: Phenotypic, genetic and reference genome characterization. *J Antimicrob Chemother.* 2016;71(11):3096–108.
5. American Academy of Pediatrics. Gonococcal Infections. In: DW Kimberlin, MT Brady, MA Jackson SL, editor. *Red Book: 2018-2021 Report of the Committee on Infectious Diseases*. 31st ed. Itasca: American Academy of Pediatrics; 2018. p. 355–65.
6. Zenilman, Jonathan, Shahmanesh M. Sexually Transmitted Infections. In: 1st ed. Sudbury: Jones and Bartlett Learning; 2012. p. 464.
7. Chan PA, Janvier M, Alexander NE, Kojic EM, Chapin K. Recommendations for the diagnosis of *Neisseria gonorrhoeae* and *Chlamydia trachomatis*, including extra-genital sites. *Med Health R I* [Internet]. 2012;95:252–4. Available from: <http://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=3763711&tool=pmcentrez&rendertype=abstract>
8. Government of Canada. Canadian Guidelines on Sexually Transmitted Infections [Internet]. 2018 [cited 2019 Jan 28]. Available from: <https://www.canada.ca/en/public-health/services/infectious-diseases/sexual-health-sexually-transmitted-infections/canadian-guidelines/sexually-transmitted-infections/canadian-guidelines-sexually-transmitted-infections-37.html>